DO NOW — RECALL

Graphs: directed and undirected

Name: _____ Class: _____ Date: _____

1. Which structure is LIFO?

2. Which structure is FIFO?

3. In an array-based linked list, what does the next pointer store?

4. In an undirected graph, if A is connected to B, is B also connected to A?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A stack.

2. Model answer

A queue.

3. Model answer

The array index of another item.

4. Model answer

Yes.

By the end, you can…

• distinguish directed and undirected graphs

• represent a graph using a 2D adjacency matrix

• add, remove and inspect edges procedurally

Vocabulary — know these terms

Recall the term — graph

Recall the term — vertex

Recall the term — edge

Recall the term — directed

Recall the term — undirected

Recall the term — adjacency matrix

Recall the term — neighbour

Recall the term — traverse

NEW LEARNING

Read, annotate and explain

Use a 2D array as an adjacency matrix

A B C
A 0 1 0
B 1 0 1
C 0 1 0

A — B — C

Undirected edges set two matrix positions. Directed edges set only the stated direction.

Create

Three vertices (A, B and C) produce a 3 by 3 matrix: one row and one column per vertex. The four-vertex 4×4 example comes later in You Do.

Add undirected edge

The matrix is symmetrical for an undirected connection.

Add directed edge

Only A→B is stored.

Remove and inspect

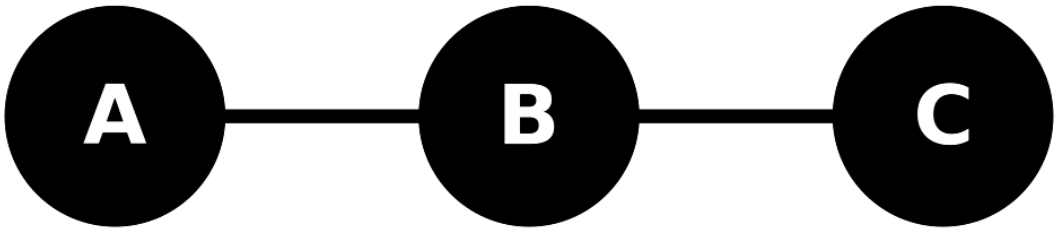
Clearing removes an edge; scanning a row finds neighbours.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Use a 2D array as an adjacency matrix



Undirected graph A connected to B connected to C

I DO — WORKED EXAMPLE

Follow the worked method

Build A—B—C

Make matrix

```
int[,] graph = new int[3,3];
```

A—B

```
graph[0,1] = 1;
graph[1,0] = 1;
```

B—C

```
graph[1,2] = 1;
graph[2,1] = 1;
```

Read neighbours of B

```
row 1 = 1 0 1
B connects to A and C
```

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Add undirected edge

Add edge 0—2.

```
graph[0,2] = ?;
graph[2,0] = ?;
```

Method / working:

2. Add directed edge

Add only 1→3.

```
graph[1,3] = ?;
```

Method / working:

3. Remove undirected edge

Remove 0—2.

```
graph[0,2] = ?;
graph[2,0] = ?;
```

Method / working:

4. Find neighbours

Inspect row 1.

```
if (graph[1,v] == ?)
    Console.WriteLine(v);
```

Method / working:

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Add undirected edge Add edge 0—2. Fill the same code CODE: graph[0,2] = ?; graph[2,0] = ?;

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 2: Add directed edge Add only 1→3.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 3: Remove undirected edge Remove 0—2.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

Guided 4: Find neighbours Inspect row 1.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Create an undirected graph
Create a 4×4 matrix and add 0—1, 0—2 and 2—3.

Question 2

List neighbours
Output every vertex directly connected to vertex 0.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Modify the graph
Add 1—3, then remove 0—2.

Question 4

Show direction matters
Create a 2×2 directed matrix with 0→1 only, then output both directional entries.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

An undirected edge normally changes two entries in an adjacency matrix.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Fill the same code

CODE:

```
graph[0,2] = 1;

graph[2,0] = 1;
```

IMPROVE: Explain one correction to your own reasoning.

Guided 2 — original CHECK

Fill the same code

CODE:

```
graph[1,3] = ?;
```

CODE:

```
graph[1,3] = 1;
```

IMPROVE: Explain one correction to your own reasoning.

Guided 3 — original CHECK

Fill the same code

CODE:

```
graph[0,2] = ?;

graph[2,0] = ?;
```

CODE:

```
graph[0,2] = 0;

graph[2,0] = 0;
```

IMPROVE: Explain one correction to your own reasoning.

Guided 4 — original CHECK

Fill the same code

CODE:

```
if (graph[1,v] == ?)
```

```
Console.WriteLine(v);
```

CODE:

```
if (graph[1,v] == 1)
Console.WriteLine(v);
```

IMPROVE: Explain one correction to your own reasoning.